AI PERSONAL STUDY ASSISTANT

MACHINE LEARNING

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MACHINE LEARNING

Concise Notes • Core Concepts • Examples • Revision Questions

B.Tech CSE (Artificial Intelligence and Machine Learning)

1. Introduction to Machine Learning

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn patterns from data and use those patterns to make predictions or decisions. Instead of writing every rule manually, a developer gives the system examples and a learning algorithm.

> **Example:** To identify spam emails, an ML model can learn from many emails labelled “spam” or “not spam.” It may learn that certain words, links, or message patterns are associated with spam.

Traditional programming vs Machine Learning

Traditional programming	Machine learning
Programmer writes explicit rules.	Algorithm learns patterns from examples.
Input + rules → output.	Training data + learning algorithm → model.
Useful when rules are clear and stable.	Useful when patterns are complex or hard to write as rules.

Basic terms

Dataset: a collection of examples. **Feature:** an input property, such as house size. **Label/target:** the value to predict, such as house price. **Algorithm:** the learning procedure. **Model:** the learned relationship used to make predictions. **Training:** fitting a model to data. **Inference:** using a trained model on new input.

Typical ML workflow

1. Define the problem and how success will be measured.
2. Collect suitable data and check quality and permissions.
3. Clean data and select useful features.
4. Split data into training, validation, and test sets as appropriate.
5. Train a simple baseline model.
6. Evaluate results and inspect errors.
7. Improve the model, then document its limitations.

Applications

Recommendation systems, fraud detection, speech recognition, medical-image support, demand forecasting, spam filtering, and image classification. ML predictions can be wrong, so the consequences of errors must be considered.

2. Types of Machine Learning

A common way to organise ML is by the kind of feedback available during learning.

A. Supervised learning

The model learns from labelled examples, where each input has a known target. It is commonly used for **classification** and **regression**.

Classification: predict a category, such as spam/not spam or disease/no disease.
Regression: predict a numerical value, such as a house price or tomorrow's temperature.

> Example: A model trained with house sizes and known sale prices may learn to estimate the price of a new house. Past data should represent the cases where the model will be used.

B. Unsupervised learning

The data has no target labels. The algorithm looks for structure, such as groups or lower-dimensional patterns.

Clustering: group similar examples, for example customers with similar purchasing behaviour.
Dimensionality reduction: represent data using fewer variables while preserving important structure, as PCA attempts to do.

C. Reinforcement learning

An agent interacts with an environment, takes actions, and receives rewards or penalties. It learns a policy that aims to maximise long-term reward. Examples include game-playing agents and some robot-control tasks.

Comparison

Type	Training signal	Example task
Supervised	Known labels or targets	Predict house prices
Unsupervised	No target labels	Group similar documents
Reinforcement	Rewards from actions	Learn a game strategy

Choosing the type

Ask: Do I have correct target answers? If yes, supervised learning may fit. If not, and I want to discover groups or structure, unsupervised learning may fit. If an agent learns by taking actions and receiving rewards, reinforcement learning may fit.

3. Important ML Algorithms

Linear regression

Used to predict a continuous numerical value. In simple linear regression, the model fits a line: ■ **= wx + b**, where x is the input, w is the learned weight, b is the bias, and ■ is the prediction. Training chooses parameters that reduce a loss, often mean squared error.

Example: Estimate a house price from its size. A line may capture a broad trend, but it may perform poorly if the relationship is nonlinear or important features are missing.

Logistic regression

Often used for binary classification. It calculates a score and maps it through a sigmoid function to obtain a value between 0 and 1, interpreted as a class probability when the model is appropriately trained and calibrated.

Decision tree

A tree makes a prediction by asking a sequence of feature-based questions. It is relatively easy to interpret, but a deep tree may memorise training examples and overfit.

K-nearest neighbours (KNN)

For a new example, KNN finds the K closest training examples and uses their labels or values to make a prediction. Feature scaling is often important because distance calculations can otherwise be dominated by large-scale features.

K-means clustering

K-means partitions examples into K clusters by repeatedly assigning points to nearby centroids and updating each centroid. The value of K must be chosen, and results can depend on scaling, initialisation, and cluster shape.

Support vector machine (SVM)

SVM seeks a decision boundary with a large margin between classes. Kernels can model certain nonlinear boundaries. Scaling features is commonly important.

Neural networks

Neural networks combine layers of weighted calculations and nonlinear activation functions. During training, backpropagation computes gradients and an optimiser updates weights to reduce the loss. Deep learning uses neural networks with multiple layers.

> Algorithm choice depends on the problem, data size and quality, interpretability needs, computing resources, and evaluation results. A more complex model is not automatically better.

4. Training, Testing, and Model Evaluation

Why split the data?

The training set is used to fit model parameters. A validation set can support model selection and hyperparameter tuning. The test set is held back for a final, less-biased estimate of performance on unseen data. Repeatedly tuning against the test set makes it less independent.

Overfitting and underfitting

Overfitting: the model learns training details or noise and performs poorly on new data. Possible responses include simplifying the model, regularisation, more representative data, cross-validation, or early stopping.
Underfitting: the model is too simple or insufficiently trained to capture the pattern. Possible responses include better features, a more suitable model, or improved training.

Common classification metrics

Accuracy = correct predictions / all predictions.
Precision = TP / (TP + FP): among predicted positives, how many were positive?
Recall = TP / (TP + FN): among actual positives, how many were found?
F1-score combines precision and recall using their harmonic mean.

> TP = true positive, TN = true negative, FP = false positive, FN = false negative. Choose metrics based on the real cost of mistakes. Accuracy can be misleading when one class is much more common than another.

Regression metrics

MAE: average absolute error; easy to interpret in the target's units.
MSE: average squared error; penalises large errors more strongly.
RMSE: square root of MSE; expressed in the target's units.

Data leakage

Data leakage happens when information unavailable at prediction time influences training or evaluation. Examples include using future information, or scaling the entire dataset before splitting it. Fit preprocessing steps on training data only, then apply them to validation/test data.

A sound evaluation checklist

- Use a split strategy appropriate to the task; time series often need chronological splits.
- Compare with a simple baseline.
- Inspect per-class results and important error types.
- Check whether test data represents real use.
- Report limitations and avoid claims beyond the evidence.

5. Practical Example: Spam Classification

Goal: predict whether an incoming message is spam or not spam. This is a supervised binary-classification problem because examples can be labelled.

Step 1 — Define inputs and labels

Inputs may include message text and carefully selected derived features. The label is spam or not spam. Use a lawful, privacy-conscious dataset and remove sensitive information when appropriate.

Step 2 — Prepare data

Check for missing or duplicated examples, examine label balance, and split the dataset into training and test sets. If many similar messages are present, take care that near-duplicates do not appear across both sets.

Step 3 — Convert text into features

A model needs numerical input. A simple approach is bag-of-words or TF-IDF, which represents text using word or term weights. Fit the text vectoriser on training data only, then transform the held-out data with that fitted vectoriser.

Step 4 — Train a baseline

A Naive Bayes classifier or logistic regression model can be a useful starting point for text classification. Train on the training data and tune choices using validation or cross-validation.

Step 5 — Evaluate

Inspect precision, recall, F1-score, and the confusion matrix. If missing spam is costly, recall may matter; if legitimate messages are frequently marked as spam, precision may matter. The best trade-off depends on user needs.

Step 6 — Review failures

Read examples of false positives and false negatives. Check whether the data is outdated, whether language or formatting differs, and whether attackers can adapt their messages. Do not rely on a single score.

Simple pipeline diagram

Labelled messages	→	Text features	→	Train model
New message	→	Same fitted features	→	Spam probability / class

Key lesson: a machine-learning project is not just training an algorithm. Data quality, evaluation design, error analysis, privacy, and maintenance are equally important.

6. Key Terms and Revision Notes

Term	Simple meaning
Feature	An input variable used by a model.
Label / target	The correct outcome the model learns to predict.
Training	Learning model parameters from examples.
Inference	Using the trained model on new input.
Hyperparameter	A setting chosen outside the model's learned parameters, such as tree depth.
Loss function	A numerical measure of how far predictions are from desired outputs.
Gradient descent	An optimisation method that updates parameters using gradients.
Regularisation	A technique that discourages overly complex fitting.
Cross-validation	Repeated train/validation splits used to estimate performance or tune models.
Class imbalance	When some labels occur much more often than others.
Data leakage	Accidentally using information that should not be available for training/evaluation.

Quick formula reminders

- Linear regression prediction: ■ = wx + b
- Accuracy = (TP + TN) / (TP + TN + FP + FN)
- Precision = TP / (TP + FP)
- Recall = TP / (TP + FN)
- F1 = 2 × (precision × recall) / (precision + recall)

Remember: formulas are useful only when you understand what each quantity means and when the metric is appropriate.

7. Exam Questions and Self-Test

Short-answer questions

1. Define Machine Learning and state two applications.
2. Differentiate traditional programming and machine learning.
3. What are features and labels?
4. Distinguish classification from regression.
5. Explain supervised, unsupervised, and reinforcement learning.
6. What is overfitting? Give two ways to reduce it.
7. Why are training and test data kept separate?
8. Define precision and recall.
9. What is data leakage? Give one example.
10. Why should a model be compared with a baseline?

Long-answer / problem questions

1. Explain the complete machine-learning workflow with a labelled diagram.
2. Compare linear regression, logistic regression, decision trees, KNN, and K-means with suitable examples.
3. Explain the confusion matrix and derive accuracy, precision, recall, and F1-score.
4. Describe how you would build and evaluate a spam classifier.
5. Explain overfitting, underfitting, and the importance of generalisation.
6. Discuss data leakage, privacy, bias, and limitations in ML applications.

Self-test: check your understanding

Question	Answer
Predicting a house price: classification or regression?	Regression
Grouping customers without known labels: which learning type?	Unsupervised learning
A model scores highly on training data but poorly on new data. What is it likely?	Overfitting
Among predicted spam messages, what does precision measure?	The proportion that really are spam
Should a text vectoriser be fitted on the test set?	No; fit on training data only

Revision plan: review the definitions, practise explaining each algorithm in your own words, solve metric calculations, and implement one small ML project. Check your official syllabus for required algorithms and depth.